Clustering in Machine Learning

Clustering is a method of organizing data by grouping similar things together into clusters. It helps to make sense of data by identifying patterns and grouping similar items .

K-Means Clustering

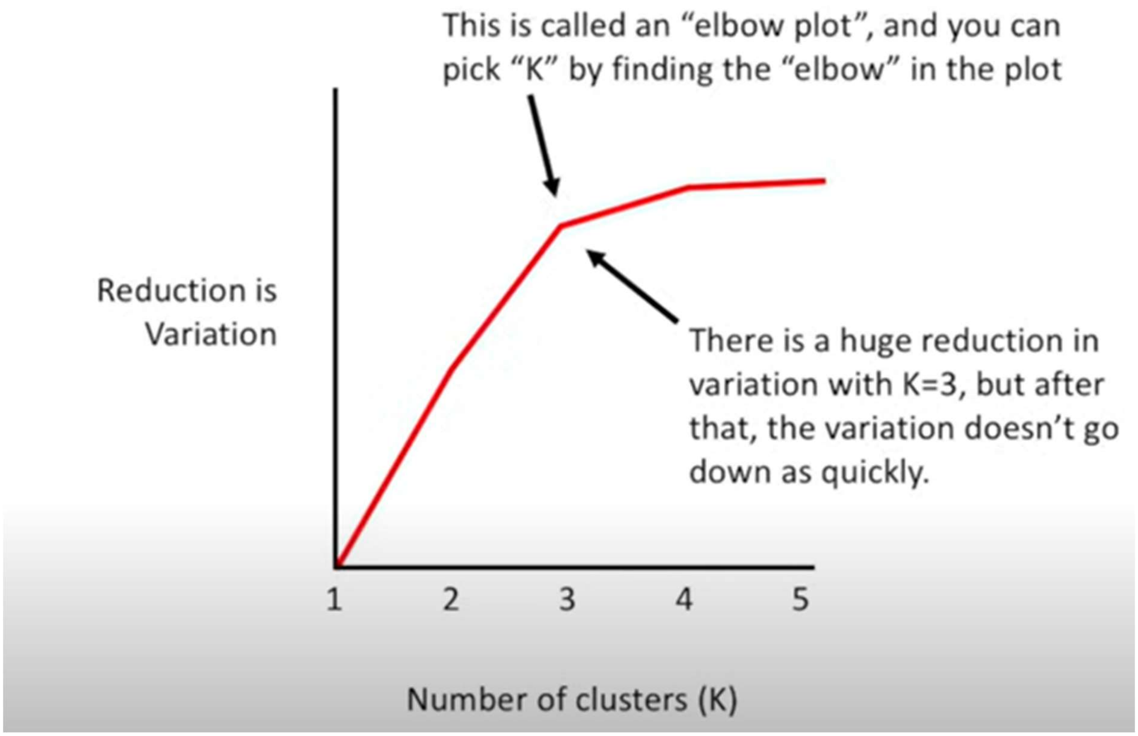
K-Means clustering is one of the most popular clustering techniques. Here's how it works:

1. **Choosing the Number of Clusters**:
   - The "K" in K-Means represents the number of clusters we want to identify in the data.
2. **Initializing Clusters**:
   - Randomly select K number of distinct data points as the initial clusters.
3. **Assigning Points to Clusters**:
   - Measure the distance between a data point and the initial clusters.
   - Assign the point to the nearest cluster.
4. **Updating Cluster Centres**:
   - Calculate the mean of each cluster and use these mean values to redefine the cluster centres.
   - Repeat the process of measuring distances and assigning points based on the updated means.
5. **Choosing the Best Clustering**:
   - The best clustering is selected by calculating the **total variance** within clusters.
   - The most accurate clustering has the smallest variance, meaning the clusters are well-defined and balanced.

Finding the Right K:

- K can be selected through **trial and error**, comparing the total variance for different values of K.
- Alternatively, an **elbow plot** can help.

- An elbow plot shows the relationship between the number of clusters and the reduction in variance.
- The "elbow" is the point where the variance reduction slows down significantly, indicating the best number of clusters.



DBSCAN (Density-Based Spatial Clustering)

DBSCAN is a clustering method designed to handle situations where K-Means struggles, such as when clusters have irregular shapes or sizes.

How DBSCAN Works:

1. **Finding Core Points**:
    - Start with the raw data and pick an initial point.
    - Draw a circle (defined by a radius) around the point.
    - Count the number of points within this circle.
2. **Defining a Core Point**:
    - If the number of points in the circle exceeds a user-defined threshold, the starting point is a **core point**, and all these points from the start of a cluster.

3. **Growing the Cluster**:
   - Points close to the initial core points are also declared as core points and added to the cluster.
   - This process continues, and the cluster keeps growing until no more core points are found.
4. **Adding Non-Core Points**:
   - Points near the cluster but not dense enough to be core points are added as **non-core points**.
   - Non-core points cannot extend the cluster further.

Advantages of DBSCAN:

- It works well with non-uniform shapes and unevenly sized clusters.
- It is reduces noise, treating outliers as separate from clusters.

CONCLUSION

Clustering, whether through K-Means or DBSCAN, helps organize and understand data. Choosing the right method depends on the dataset's characteristics and the clustering requirements.